

NUH Paediatric Trauma Protocol

This protocol addresses the following:

- 1. Types of trauma activation and what this entails
- 2. Location of paediatric trauma resuscitation
- 3. Roles of trauma team members
- 4. Essentials of paramedic handover
- 5. Basic trauma template
- 6. Massive transfusion protocol in paediatric patients
- 7. Paediatric trauma resuscitation drugs

1. Types of trauma activation and what this entails

Paediatric Trauma Standby

- ED team calls 24000, states "Paeds trauma standby" and location of standby
- Trauma team (GS Registrar and MO on call, Paediatric Surgery Registrar and MO on call) notified by SMS, will **not** come down unless activation happens.

Paediatric Trauma Activation (also known as **Code trauma Tier 2**)

- ED team calls 24000, states "Paeds trauma activation" and location of resus
- Trauma team (GS Registrar and MO on call, Paediatric Surgery Registrar and MO on call) notified by SMS and phone call, will come down to where patient is

Code trauma Tier 1 Activation

- ED team calls 24000, states "code trauma tier 1 - paediatrics" and location of resus
- Trauma patients for whom **MTP is activated/urgent blood transfusion** is required
- This aims to prepare various teams and locations for transfer of an unstable trauma patient out of Emergency Department to Operating OT/IR department as soon as possible
- Criteria for activation:

- 2 or more of the following:
 - ✓ Hypotension
 - ✓ Tachycardia
 - ✓ Presence of penetrating trauma
 - ✓ FAST positive
- Poor response to initial resuscitation due to suspected active haemorrhage

- These teams will be notified by SMS and phone call:

GS Registrar and MO on call
Trauma consultant (weekday office hours)
Paediatric Surgery Registrar and MO on call
Anaesthesia Registrar and Consultant running Emergency OT
Neurosurgery Registrar
Orthopaedic Surgery Registrar
Interventional radiology (IR) nurse on call
OT nurse
Radiologist on call
EMD on duty radiographer

- Paediatrics ICU on call needs to be informed by CE team for bed arrangement

2. Location of paediatric trauma resuscitation

Location of major paediatric trauma resuscitation – Adult ED trolley 5 or 6

If patient is dropped in to Children's Emergency without prior notice, and

- a) Is deemed unstable (hypotensive), resuscitate in Children's Emergency first. Inform adult ED senior to help eg: FAST scan.
- b) Is deemed as major trauma and is not hypotensive, move to adult ED trolley 5 or 6 after informing ED resus senior to ensure space availability.
- c) Is deemed as minor trauma, continue management in Children's Emergency.

3. Roles of trauma team members

Trauma team leader – Adult ED senior

Doctor 1 – Airway doctor: in charge of airway, breathing and disability (A+B+D)

Doctor 2 – Circulation doctor: in charge of circulation and exposure (C+E)

Doctor 3 – Trauma team leader (TTL): adult ED senior, +/- CE senior co-leading

Nurse 1 – Airway nurse

Nurse 2 – Circulation/medication nurse

Nurse 3 – Scribe nurse

MTP nurse – Obtains PCT from ED blood fridge, informs blood bank blood products required.

4. Essentials of paramedic handover

Important information from paramedics' handover (**MIST**)

-**M**echanism of injury eg: height of fall, speed of travel, casualty on scene

-**I**njuries sustained

-Vital **S**igns on scene – HR, BP, SpO2, GCS

-**T**reatment rendered

5. Basic trauma template

General trauma principles:

-Attend to all catastrophic bleeds first

-Primary survey: Airway + C spine, Breathing, Circulation, Disability, Exposure

-Primary series: CXR, Pelvic XR, eFAST

-Secondary survey: Head-to-toe examination

-Secondary series: Scans

6. Massive transfusion protocol in paediatric patients

Blood Transfusion

Activation of Massive Transfusion Protocol (MTP)

- Criteria – Age > 12 years old OR Weight > 40kg
- Consists of 4 units of O+ Blood type PCT, 4 units of FFP, 1 pooled platelet

Updated by Dr Ivy Ang, May 2024.

- EMD blood fridge
 - Consists of 4 units of O+ PCT, of which 1 unit is leucocyte-reduced O+ PCT.
 - Traditionally, this blood resource is reserved for patients for whom MTP is activated
 - **For paediatric trauma patients exhibiting signs of haemorrhagic shock requiring urgent blood transfusion but do not qualify for MTP, TTL and/or CE senior can ask for ED blood fridge packed cells to be given to patient.** This is to facilitate timely blood transfusion as per tenets of haemostatic resuscitation.
 - MTP ED nurse withdraws blood products one unit at a time. Nurse 2 (medication nurse) will be in charge of administering the blood products.
 - Contraindications to use of O+ PCT from ED Blood fridge:
 - ✗ Female patients of Caucasian / Indian descent and of child bearing age with unknown blood type- they will require O- PCT
 - ✗ Patients known to be Rhesus negative
 - One unit of Leucocyte reduced packed cell is available in the ED blood fridge which both paediatric and adult patients can utilize as part of MTP activation.
 - Leucocyte reduced packed cells should be given in:
 - ✓ All children <1 year old
 - ✓ Those at risk of febrile haemolytic reactions
 - ✓ Those at risk of CMV transmission (immunocompromised patients)

Types of blood products from blood bank:

For Red Blood Cells				
Type of Crossmatch	Time from ordering or if urgent, calling BTS to first 2 PCT ready for collection*	GXM required?	ABO group	Transfusion Reaction Risk
Uncrossmatch	5 – 10 mins	No. Can be issued before GXM**	O+ for males O- for Indian and Caucasian females <50 y	5% risk of incompatibility 0.5% risk of haemolytic reaction
Rapid Match	15min from time GXM sample arrives in BTS	Yes	ABO specific No antibody screen	5% risk of incompatibility 0.5% risk of haemolytic reaction
Full Crossmatch	15min (valid GXM in last 3 days, without antibody) 45min from the time GXM sample arrives in BTS 2-4 hours (with antibody)	Yes	ABO specific Antibody screen done	Lowest risk of haemolytic reaction

For urgent blood products that can only be supplied by blood bank, send a doctor/nurse up to blood bank with patient's sticker for identification purposes.

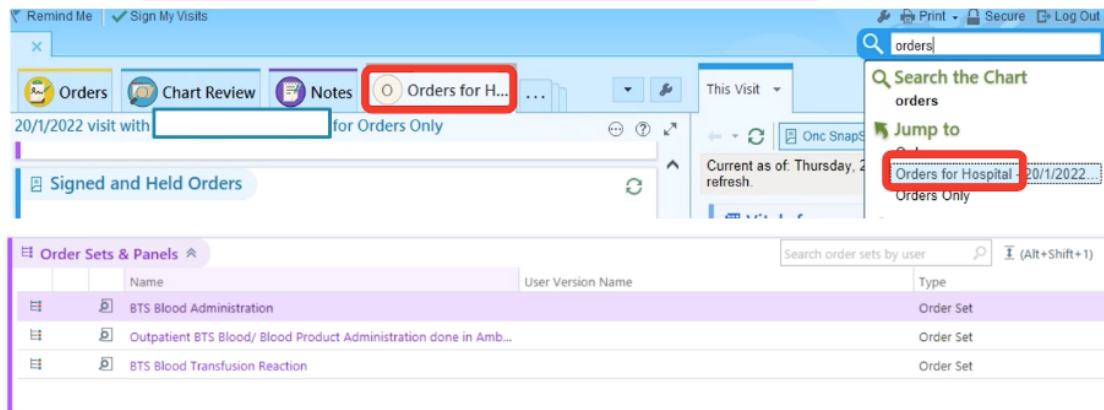
In cases of haemorrhagic shock, blood products is be run as a fluid bolus, and not over 4 hours. This is to replace the loss of circulating volume which has resulted in haemodynamic instability.

How to order

- Go to orders
- Key in BTS Blood administration

Proceed to "Order for Hospital" and open the patient tab
Under Order Set, type in "BTS Blood Administration" and click on the order set.

Tip: Do not order in "Add Orders"



The screenshot displays a medical software interface. At the top, there are tabs for 'Orders', 'Chart Review', 'Notes', and 'Orders for H...'. The 'Orders for H...' tab is selected and highlighted with a red box. Below the tabs, there is a search bar with the text 'orders' and a 'Jump to' button. A red box highlights the 'Orders for Hospital' link in the 'Jump to' section. Below the search bar, there is a table titled 'Order Sets & Panels' with columns for 'Name', 'User Version Name', and 'Type'. The table lists three order sets: 'BTS Blood Administration', 'Outpatient BTS Blood/ Blood Product Administration done in Amb...', and 'BTS Blood Transfusion Reaction'. The first row is highlighted in purple.

Name	User Version Name	Type
BTS Blood Administration		Order Set
Outpatient BTS Blood/ Blood Product Administration done in Amb...		Order Set
BTS Blood Transfusion Reaction		Order Set

7. Paediatric trauma resuscitation drugs

For RSI

Medication	Dose (max dose)	Notes
Etomidate	0.3mg/kg (max 20mg)	BP stable Has no analgesia
Midazolam	0.1mg/kg (max 10mg)	Can cause hypotension Has no analgesia
Fentanyl	0.5-1mcg/kg for analgesia 1-3mcg/kg for RSI (max 100mcg)	
Ketamine	1mg/kg (max 100mg)	Has both sedation and analgesia
Suxamethonium	1-2mg/kg (max 150mg)	Short acting paralytic agent
Rocuronium	1mg/kg (max 100mg)	Longer acting paralytic agent

Other medications

Medication	Dose (max dose)	Notes
Tranexamic acid	15mg/kg (max 1g) over 15 mins	Indicated for all trauma patients who are bleeding or at risk of significant haemorrhage. Give early (within 3 hours following an injury).
Hypertonic 3% saline	3-5ml/kg	
Keppra	10mg/kg as prophylaxis (max 500mg)	